



ColourSpace Remote Control Protocol

It is possible to remote control the measurement of patches within ColourSpace.

To do this, you must have an active profiling window open, with a connected probe, and with the Remote Control option set to servant mode.

In order to establish a connection, you must first open a TCP connection to the ColourSpace host, using the port number you have set in the ColourSpace GUI.

Once connected, all communications are performed using structured XML.

Each command (and return message) should be preceded by a single integer value, giving the total number of following characters.

An example implementation of a remote control program is supplied as source code in C++ within the ColourSpace distribution, as well as within the download .zip file that contains this document.

The available commands and returns are shown as follows:

Initialisation Command

The init command clears any existing data and then initialises the profile window. The init command can have either of the following: profile, quickProfile, quickGreyProfile, which determines the type of profile you wish to create.

Use profile as a default.

It is the user's responsibility to send a meaningful set of patches to measure to ensure the profile is valid.

```
<?xml version="1.0" encoding="UTF-8" ?>
  <CS_RMC version=1>
    <command>
      init profile
    </command>
  </CS_RMC>
```



Measure Patch

The measurement request sends the data point to be measured. It is comprised of 3 integer values, between 0-255 that represent the patch colour to be measured.

```
<?xml version="1.0" encoding="UTF-8" ?>
<CS_RMC version=1>
  <measurement>
    <red>255</red>
    <green>255</green>
    <blue>255</blue>
    <bits>8</bits>          <!-- optional, 8, 10,12,14,16; default 8 -->
    <mode>FULL</mode>      <!-- optional, FULL, LEGAL, EXTENDED; default FULL -->
  </measurement>
</CS_RMC>
```

Patch Returned

The result data returned is shown as follows.

```
<?xml version="1.0" encoding="UTF-8" ?>
<CS_RMC version="1">
  <result>
    <red>255</red>
    <green>255</green>
    <blue>255</blue>
    <x>0.313656418979</x>
    <y>0.329456868400</y>
    <Y>180.954114250711</Y>
    <bits>8</bits>
    <mode>FULL</mode>
  </result>
</CS_RMC>
```



Calculate Metrics

The metric request is used to receive the calculated values as displayed in the GUI window. The data sent can be parsed from the result of a successful measurement request.

```
<?xml version="1.0" encoding="UTF-8" ?>
<CS_RMC version=1>
  <metric>
    <red>255</red>
    <green>255</green>
    <blue>255</blue>
    <x>0.313656</x>
    <y>0.329457</y>
    <Y>180.954</Y>
    <bits>8</bits>          <!-- optional, 8, 10,12,14,16; default 8 -->
    <mode>FULL</mode>      <!-- optional, FULL, LEGAL, EXTENDED; default FULL -->
  </metric>
</CS_RMC>
```

Metric Returned

The metric request returns the calculated values as displayed in the GUI window. It should be noted that the received values are based on the current GUI settings. The values tY, tx, tz are the ideal values for the current standard.

```
<?xml version="1.0" encoding="UTF-8" ?>
<CS_RMC version="1">
  <metric>
    <red>255</red>
    <green>255</green>
    <blue>255</blue>
    <x>0.313656000000</x>
    <y>0.329457000000</y>
    <Y>180.954000000000</Y>
    <tx>0.312700000000</tx>
    <ty>0.329000000000</ty>
    <tY>67.590800000000</tY>
    <de76>45.075661729331</de76>
    <de20>21.619778009663</de20>
    <deltP>71.574654113750</deltP>
    <minY>0.039000000000</minY>
    <maxY>67.590800000000</maxY>
    <std>Rec709</std>
    <bits>8</bits>
    <mode>FULL</mode>
  </metric>
</CS_RMC>
```



Erase Command

The erase command deletes all current measurements, retaining the current profile type.

```
<?xml version="1.0" encoding="UTF-8" ?>
  <CS_RMC version=1>
    <command>
      erase
    </command>
  </CS_RMC>
```

Cal Window Command

The calWindow command makes the calibration patch window visible (on) or invisible (off).

```
<?xml version="1.0" encoding="UTF-8" ?>
  <CS_RMC version=1>
    <command>
      calWindow on          <!-- option on or off-->
    </command>
  </CS_RMC>
```

Set Stabilisation

The stabilisation request controls the stabilisation settings. Stabilisation can be enabled or disabled. You can optionally set any of the other parameters. If they are not present in the request, they remain unchanged.

```
<?xml version="1.0" encoding="UTF-8" ?>
<CS_RMC version=1>
  <stabilisation>
    <active>1</active>          <!-- 1 on, 0 off. If not present, no change -->
    <fw>1</fw>                 <!-- 1 enabled, 0 disabled. If not present, no change -->
    <time>1.25</time>          <!-- stabilisation time in seconds. If not present, no change -->
    <red>255</red>             <!-- stabilisation colour. If not present, no change -->
    <green>255</green>
    <blue>255</blue>
  </stabilisation>
</CS_RMC>
```



Set Patch Size

The patchSize request controls the patch size settings. The value of diag must match the numeric percentage value in the ColourSpace GUI selector. A value of 0 means 'custom' and a value of '100' means FW *full width). The values of area, posX and posY only take effect if diag id 0 (custom). If any parameter is not present in the request, they their value remain unchanged.

```
<?xml version="1.0" encoding="UTF-8" ?>
<CS_RMC version=1>
  <patchSize>
    <diag>55</ diag>          <!-- integer percentage %. MUST match CS GUI. 0 custom -->,
                                <!-- 100 FW If not present, no change -->
    <area>90.0</area>         <!-- 0 – 100 percent. If not present, no change -->
    <posX>10</posZ>          <!-- 0 – 100 percent. If not present, no change -->
    <posY>10</posY>          <!-- 0 – 100 percent. If not present, no change -->
                                <!-- area posX, posY only work if diag is 0 (custom) -->
  </patchSize>
</CS_RMC>
```

Set BG Colour

The setBGcolour request controls the background patch colour.

```
<?xml version="1.0" encoding="UTF-8" ?>
<CS_RMC version=1>
  <bgColour>
    <red>255</red>           <!-- bg colour colour. If not present, no change -->
    <green>255</green>
    <blue>255</blue>
    <bits>8</bits>           <!-- optional, 8, 10,12,14,16; default 8 -->
  </bgColour>
</CS_RMC>
```